

RS Vorlesung 8

HA) 1)

$$G_0(s) = \frac{K(s+1)}{s^2(s+9)}$$

$$1 + G_0 = 0 \Rightarrow 0$$

$$s^2(s+9) + K(s+1) = 0 \Rightarrow 0$$

$$s^3 + 9s^2 + Ks + K = 0$$

$$\begin{aligned} (s+3)^3 &= (s^2 + 6s + 9)(s+3) \\ &= (s^3 + 9s^2 + 27s + 27) \end{aligned}$$

$\downarrow$ $\downarrow$
 K K

HA) 2) $N(s) = 1 + G_R G_S = 0$

$$= (1+s)(1+1,5s) + 4K = 0$$

$$1,5s^2 + 2,5s + (1+4K) = 0$$

$$a_2 s^2 + a_1 s + a_0 = 0$$

$$\begin{aligned} a_2 &> 0 \quad \checkmark \quad \text{stabil} \\ a_1 &> 0 \quad \checkmark \end{aligned}$$

$$1+4K = a_0 > 0$$

$$\text{wenn } K > -\frac{1}{4} \quad \checkmark \quad \text{stabil}$$

HA) 3) $\frac{Y}{W} = \frac{G_S + G_R}{1 + G_S R_R}$

$\underbrace{1 + G_S R_R}_{1 + G_0}$

$$N(s) \cancel{N(s)} = 1 + G_0 = 0$$

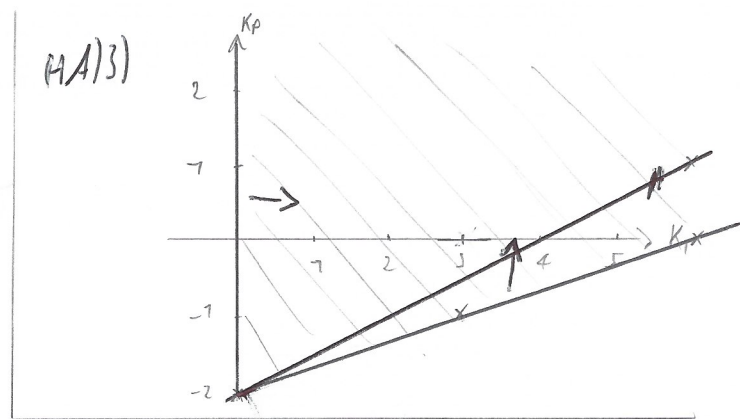
$$1 + \frac{K_p}{(s+1)(s+2)} + \frac{K_i}{(s+1)(s+2)s} = 0$$

$$s^3 + 3s^2 + 2s + sK_p + K_i = 0$$

$$\begin{aligned} a_0 &= K_i & K_i > 0 \quad \checkmark \quad \text{stabil} \\ a_1 &= K_p + 2 & K_p > 2 \quad \checkmark \quad \text{stabil} \\ a_2 &= 3 & \checkmark \quad \text{stabil} \\ a_3 &= 1 & \checkmark \end{aligned}$$

$$a_1 > \frac{K_i}{3} \quad (a_2 a_1 - a_0 a_3 > 0)$$

$$K_p > \frac{K_i}{3} - 2 \quad \checkmark \quad \text{stabil}$$



K muss 27 sein, damit die Regelkreispole bei -3 liegen.